

Customer Shopping Behavior

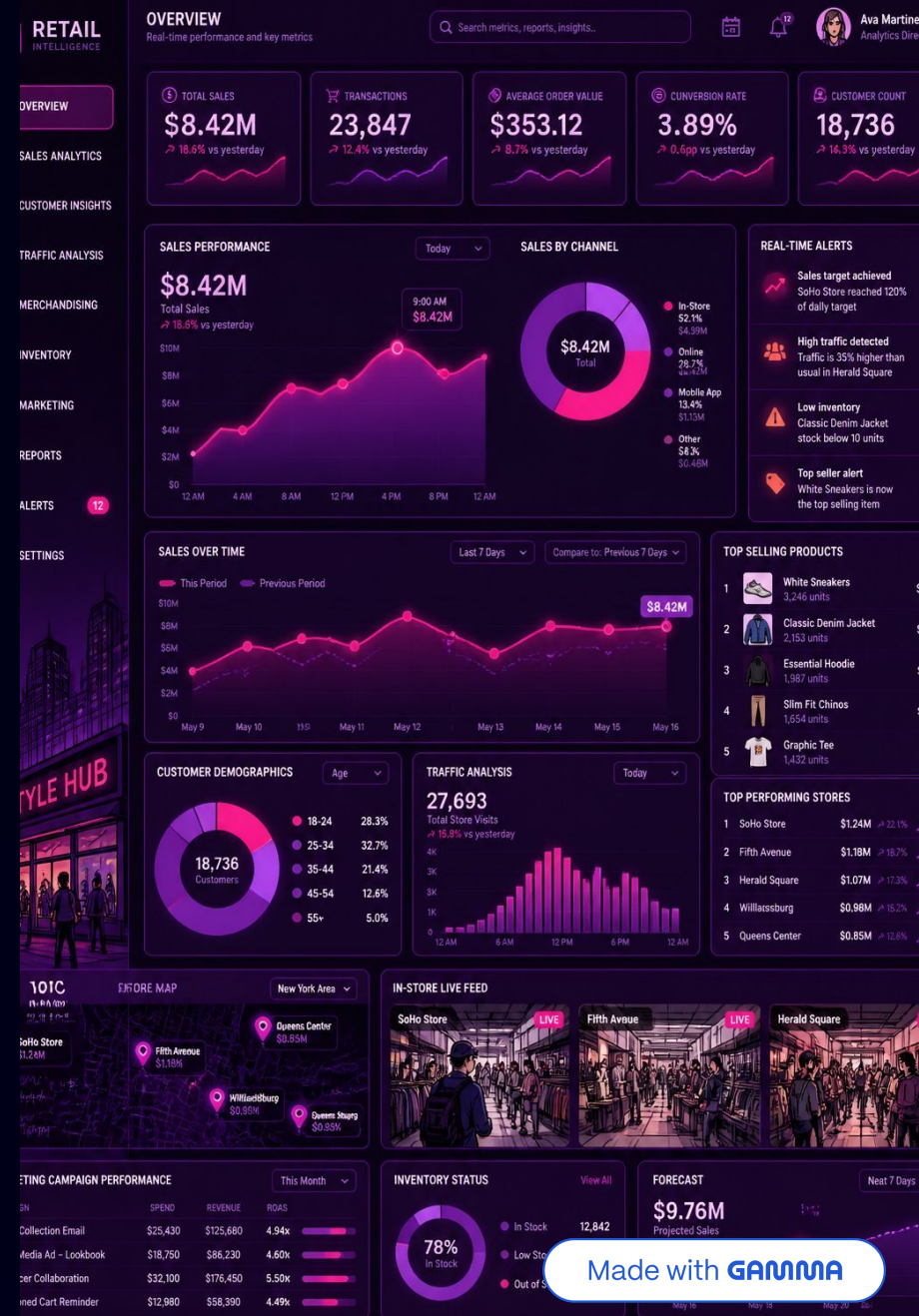
End-to-End Data Analytics Project — 3,900 transactions analyzed across the full pipeline: Python data cleaning, SQL Server business analysis, and Power BI visualization.

PYTHON (PANDAS)

SQL SERVER

POWER BI

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Project Overview

A retail dataset of **3,900 transactions** across **18 attributes** — demographics, product categories, purchase amounts, ratings, subscriptions, and shipping — processed through a complete analytics pipeline.

1

Python

Load, clean & engineer features

2

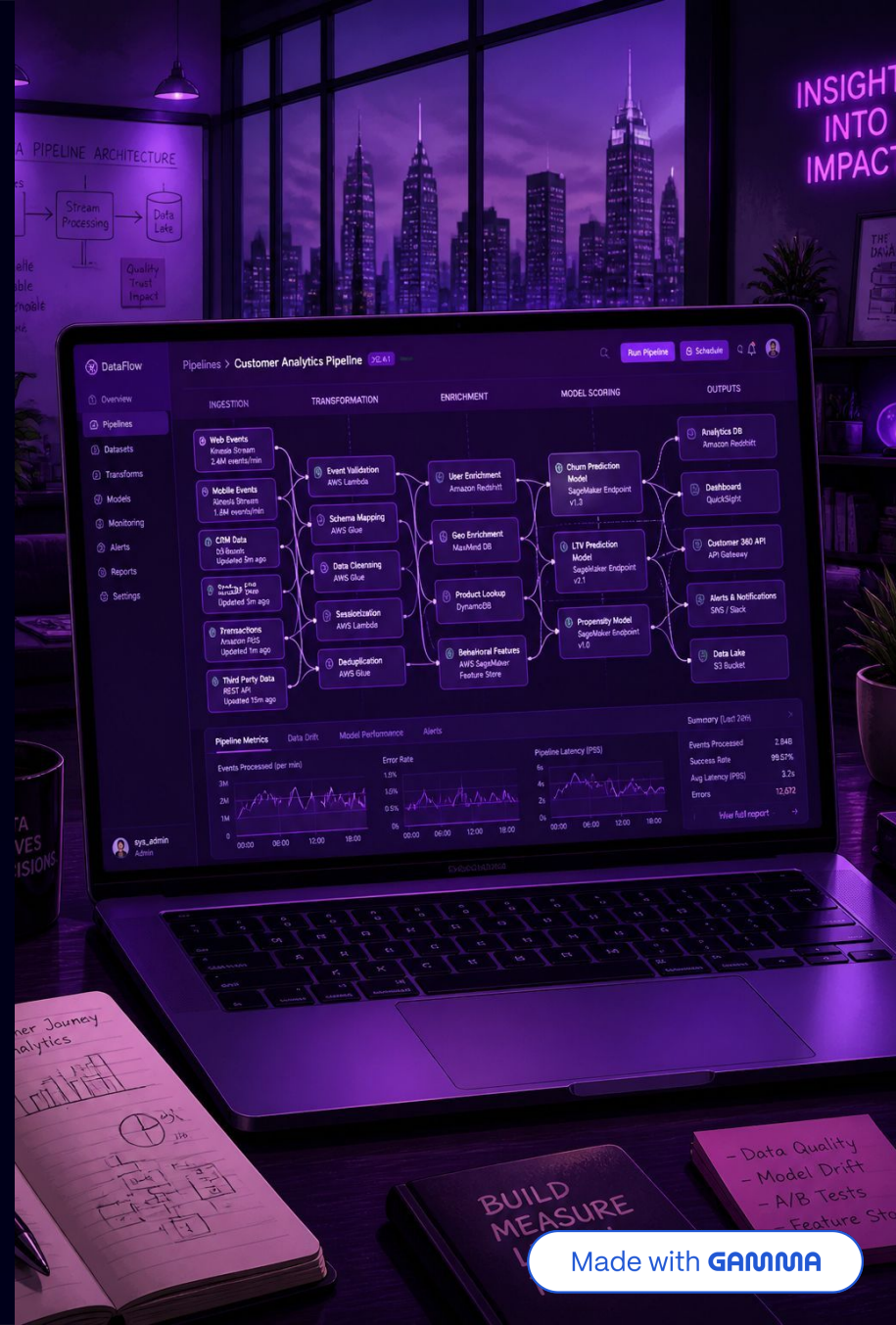
SQL Server

Store data & answer 10 business questions

3

Power BI

Interactive stakeholder dashboard



Dataset Structure

3,900 rows × 18 columns — one row per customer transaction.

Customer ID	Age	Gender	Item Purchase	Category	Purchase Amount (US)	Location	Size	Color	Season	Review Rating	Subscription Status	Shipping Type	Discount Applied	Promo Code Used	Previous Purchase
1	55	Male	Blouse	Clothing	53	Kentucky	L	Gray	Winter	3.1	Yes	Express	Yes	Yes	
2	19	Male	Sweater	Clothing	64	Maine	L	Maroon	Winter	3.1	Yes	Express	Yes	Yes	
3	50	Male	Jeans	Clothing	73	Massachusetts	S	Maroon	Spring	3.1	Yes	Free Shipping	Yes	Yes	
4	21	Male	Sandals	Footwear	90	Rhode Island	M	Maroon	Spring	3.5	Yes	Next Day Air	Yes	Yes	
5	45	Male	Blouse	Clothing	49	Oregon	M	Turquoise	Spring	2.7	Yes	Free Shipping	Yes	Yes	
6	46	Male	Sneakers	Footwear	20	Wyoming	M	White	Summer	2.9	Yes	Standard	Yes	Yes	
7	63	Male	Shirt	Clothing	85	Montana	M	Gray	Fall	3.2	Yes	Free Shipping	Yes	Yes	
8	27	Male	Shorts	Clothing	34	Louisiana	L	Charcoal	Winter	3.2	Yes	Free Shipping	Yes	Yes	
9	26	Male	Coat	Outerwear	97	West Virginia	L	Silver	Summer	2.6	Yes	Express	Yes	Yes	
10	57	Male	Handbag	Accessories	31	Missouri	M	Pink	Spring	4.8	Yes	2-Day Shipping	Yes	Yes	
11	53	Male	Shoes	Footwear	34	Arkansas	L	Purple	Fall	4.1	Yes	Store Pickup	Yes	Yes	
12	30	Male	Shorts	Clothing	68	Hawaii	S	Olive	Winter	4.9	Yes	Store Pickup	Yes	Yes	
13	61	Male	Coat	Outerwear	72	Delaware	M	Gold	Winter	4.5	Yes	Express	Yes	Yes	
14	65	Male	Dress	Clothing	51	New Hampshire	M	Violet	Spring	4.7	Yes	Express	Yes	Yes	
15	64	Male	Coat	Outerwear	53	New York	L	Teal	Winter	4.7	Yes	Free Shipping	Yes	Yes	
16	64	Male	Skirt	Clothing	81	Rhode Island	M	Teal	Winter	2.8	Yes	Store Pickup	Yes	Yes	
17	25	Male	Sunglasses	Accessories	36	Alabama	S	Gray	Spring	4.1	Yes	Next Day Air	Yes	Yes	
18	53	Male	Dress	Clothing	38	Mississippi	XL	Lavender	Winter	4.7	Yes	2-Day Shipping	Yes	Yes	
19	52	Male	Sweater	Clothing	46	Montana	S	Black	Summer	4.5	Yes	Free Shipping	Yes	Yes	
20	66	Male	Pants	Clothing	90	Rhode Island	M	Green	Summer	3.3	Yes	Standard	Yes	Yes	
21	21	Male	Pants	Clothing	51	Louisiana	M	Black	Winter	2.8	Yes	Express	Yes	Yes	
22	31	Male	Pants	Clothing	62	North Carolina	M	Charcoal	Winter	4.1	Yes	Store Pickup	Yes	Yes	
23	56	Male	Pants	Clothing	37	California	M	Peach	Summer	3.2	Yes	Store Pickup	Yes	Yes	
24	31	Male	Pants	Clothing	88	Oklahoma	XL	White	Winter	4.4	Yes	Express	Yes	Yes	

Demographics

Age, Gender, Location

Product

Category, Item, Size, Color, Season

Transaction

Purchase Amount, Payment Method, Shipping Type

Behavior

Review Rating, Subscription, Discount, Promo Code, Frequency

Only **Review Rating** had missing values: 37 out of 3,900 (~0.95%). All other columns complete.

Data Cleaning

Category-Aware Imputation

Missing Review Ratings (37 values) were filled using the **median within each product category** – preserving natural rating distributions rather than applying a blanket global average.

Column Standardization

All column names converted to **lowercase snake_case** (e.g., `purchase_amount_usd` → `purchase_amount`) for clean, error-free SQL compatibility.



Feature Engineering

age_group

Quantile-based binning (`pd.qcut`) into 4 equal-sized segments: **Young Adult**, **Adult**, **Middle-aged**, **Senior** — ensuring fair, balanced comparisons.

purchase_frequency_days

Categorical frequency (Weekly, Monthly, etc.) mapped to numeric days-between-purchases for quantitative cadence analysis.

promo_code_used → Dropped

100% identical to `discount_applied` across all 3,900 rows — fully redundant, removed to simplify schema.

Exploratory Data Analysis

3,900

Total Transactions

\$59.76

Avg. Purchase

Range: \$20–\$100

3.75

Avg. Rating

Out of 5 stars

25.35

Avg. Prev. Purchases

Range: 1–50

Customer ages span 18–70, supporting quartile-based segmentation. The wide range of previous purchases (1–50) forms the basis for New / Returning / Loyal customer tiers in SQL.

SQL Business Questions

10 T-SQL queries answered across four analytical themes:

01

Revenue Analysis

Total revenue by gender (Q1), subscribers vs. non-subscribers (Q5), revenue by age group (Q10)

02

Customer Segmentation

New / Returning / Loyal tiers via CTE + CASE logic (Q7); repeat buyers vs. subscription (Q9)

03

Discount Behavior

Discount users spending above average (Q2); products most dependent on discounting (Q6)

04

Product Performance

Top-rated products (Q3), shipping type vs. spend (Q4), top 3 per category via window function (Q8)

Power BI Dashboard



Dashboard Components

- **KPI Cards:** 3.90K customers, \$233.08K revenue, \$59.76 avg. purchase, 3.75 avg. rating
- **Donut Chart:** Subscription status — 27% subscribers vs. 73% non-subscribers
- **Bar Charts:** Revenue & sales by Category and Age Group
- **Slicers:** Filter by Subscription, Gender, Category, Shipping Type

❏ KPIs match Python EDA exactly — confirming full pipeline consistency from raw data to dashboard.

Key Insights

Clothing Dominates

Top category by both revenue and sales volume — the primary business driver.

Young Adults Lead Revenue

Highest-spending age segment; equal group sizes confirm this reflects real behavior, not sample bias.

Subscription Gap

Only 27% subscribe — a high-value minority worth targeting for conversion.

Pipeline Integrity

Python → SQL → Power BI end-to-end lifecycle validated with consistent metrics throughout.



Recommendations

1

Convert Non-Subscribers

Target high-frequency non-subscribers (5+ purchases) with trial offers or first-order discounts to drive recurring revenue.

2

Diversify Beyond Clothing

Bundle Outerwear & Footwear with top Clothing items; run category-specific promos to reduce single-category dependency.

3

Focus & Test Age Segments

Prioritize Young Adult & Middle-aged channels for near-term ROI; run win-back tests for Adult/Senior segments to assess product-fit gaps.